Hadronic Secondary-Interaction Systematics for the DUNE ND-LAr Demonstrator (2×2) with `Geant4Reweight`

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This note documents the implementation of hadronic secondary-interaction (SI) systematic uncertainties for the DUNE ND-LAr Demonstrator (2x2) using Geant4Reweight (G4RW). To enable this, the EDepSim persistency manager was patched to retain full Geant4 trajectories, exclusive-channel fraction files were produced for charged pions (both π^+ and π^-), protons, and neutrons, and the G4RW framework was extended to support nucleon reweighting. Per-track SI weights were then generated for pions, protons, and neutrons and matched to CAF observables using truth-level identifiers. The resulting weights were validated at the single-track and distribution levels and propagated to both reconstructed and truth-level observables, including proton track-length and charged-track multiplicity distributions, alongside GENIE and detector systematic uncertainties. This establishes a reproducible SI systematic framework for 2x2 analyses and provides an important ingredient for upcoming physics measurements. This approach will also work for DUNE ND-LAr.

I. INTRODUCTION

Neutrino interactions in the DUNE ND-LAr 2×2 prototype produce hadrons — predominantly protons and charged pions — which must traverse the liquid argon before they are reconstructed. Along the way they can scatter elastically, be absorbed, charge-exchange, or produce additional hadrons. These processes are collectively referred to as secondary interactions (SI), to distinguish them from final-state interactions occurring inside the struck nucleus. Secondary interactions change the number, type, and kinematics of the particles that reach the reconstruction, so the cross sections that govern them enter every hadronic observable the 2×2 measures. Figure 1 gives a representative example: a simulated neutrino interaction producing a muon, a proton, and a pion from a common vertex, reconstructed with SPINE [1] and shown alongside the truth view. The two hadrons in this event are exactly the objects whose re-interaction probability G4RW is being used to vary.

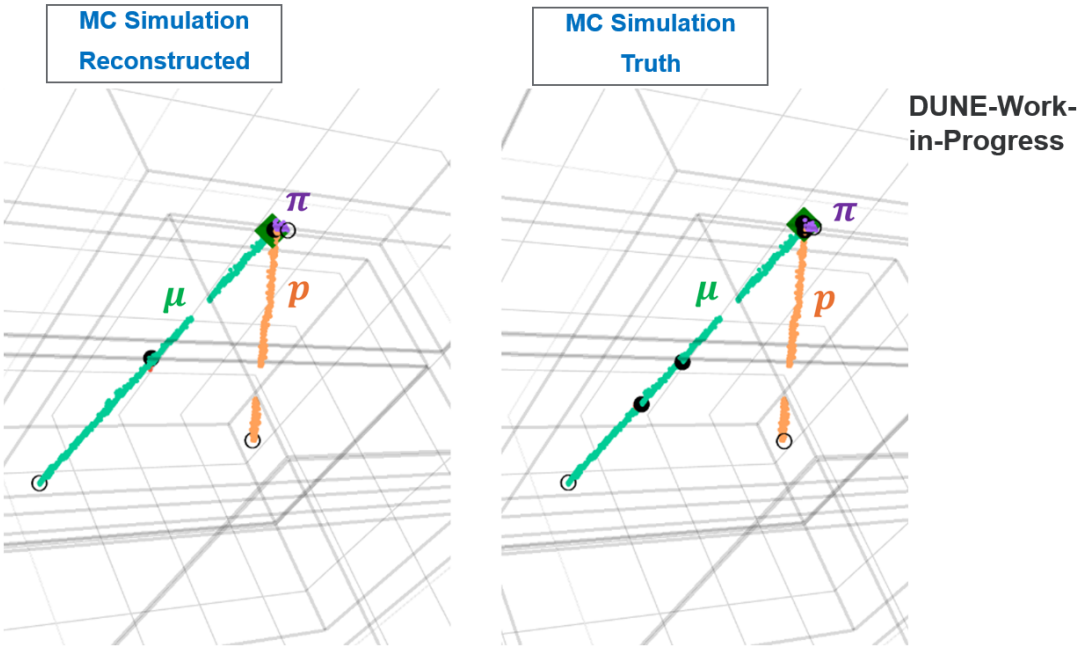


FIG. 1. Simulated neutrino interaction in the 2×2, shown in the reconstructed (left) and truth (right) views. A muon, a proton, and a charged pion emerge from a common vertex.

Geant4 determines the re-interaction probability using cross sections looked up from its own internal tables, deciding at each step whether the particle survives or interacts. Geant4Reweight (G4RW) [2] assesses the associated uncertainty

by varying those same cross sections and propagating the resulting shift to physics observables.

That SI uncertainty is one of several sources affecting the 2×2 measurement. A related but separate source is uncertainty in the neutrino interaction itself, the cross section for producing a given final state at the vertex, which is evaluated with the `nusystematics` framework [3] rather than G4RW. The direct way to assess the SI uncertainty would be to rerun the full `Geant4` [4] simulation with modified hadronic cross sections, once per variation. That is prohibitively expensive. G4RW avoids it: for a particle whose trajectory has already been simulated, it computes analytically how much more or less probable that trajectory would have been under a different cross section, and assigns the ratio as a per-track weight. Applying the weight when filling a histogram gives the shifted prediction without a new simulation pass.

Using G4RW on the 2×2 required work in four areas: regenerating the EDepSim output to retain full trajectories, extending the framework to species beyond π^+, implementing and validating the interaction-step weight formula, and connecting the weights to reconstructed quantities in the CAF files. These areas structure the note: Section II describes the weight calculation itself, Section III through Section V cover the four areas above in turn, Section VI presents a continuous parameterization of the weight versus true track length, Section VII covers matching weights to reconstructed and truth-level particles and building the systematic band, and Section VIII applies these to two example analyses. Section IX lists software and file locations on NERSC for reproducibility, and Section X gives the current state of the work.

The simulation samples used here are MiniRun5 [5] for the EDepSim trajectories and MiniRun6.5 [6] for the reconstructed CAF files. Earlier presentations of this work are recorded in Refs. [7–9], and the proton distributions to which the uncertainty is currently applied are documented in Ref. [10].

II. THE `GEANT4REWEIGHT` FRAMEWORK

G4RW [2, 11, 12] takes a trajectory that `Geant4` has already simulated and asks how probable that same trajectory would have been under a modified hadronic cross section. The ratio of the two probabilities is the weight:

$$w = \frac{P_{\text{new}}(\text{trajectory} \,|\, \sigma_{\text{new}})}{P_{G4}(\text{trajectory} \,|\, \sigma_{G4})}. \tag{1}$$

Nothing is re-simulated. The positions and momenta of the particle at each transport step are read back from the simulation output, and the cross sections at those momenta are looked up from `Geant4`'s internal tables.

A note on units. Throughout this section, and the G4RW paper [2], the symbol σ denotes an inverse mean free path with units of cm^{-1}, not a cross section in barns; the number density and molar mass factors are absorbed into it. This is the convention the code follows as well.

A. Per-Step Weight Calculation

A track propagating through argon is reweighted as a product of two pieces. At every step where the particle does *not* interact, the weight is the ratio of survival probabilities under the varied and nominal cross sections:

$$W_S = \prod_i \frac{\exp\left[-\Delta L_i \left(\sigma'_{I,i} + \sigma'_{E,i}\right)\right]}{\exp[-\Delta L_i \left(\sigma_{I,i} + \sigma_{E,i}\right)]}, \tag{2}$$

where ΔL_i is the length of step i, and σ_I and σ_E are the inelastic and elastic terms, primed quantities denoting the varied values.

At the final step, and only if the track does interact hadronically, the last-step weight is computed by:

$$W_I = f_I \left(\frac{\sigma_i}{\sigma'_i}\right) \left(\frac{1 - e^{-\Delta L_i \sigma'_i}}{1 - e^{-\Delta L_i \sigma_i}}\right). \tag{3}$$

Here $\sigma_i = \sigma_I + \sigma_E$ is the total (inelastic plus elastic) cross section at the interacting step. The first factor, f_I, is the cross-section scale factor. That is the factor scale up or down to vary the hadron SI cross section. The remaining two factors account for how the overall probability of interacting at that step changes under the varied cross section.

The total per-track weight is the product,

$$W = W_S \times W_I. \tag{4}$$

The two terms always pull in opposite directions, but which one dominates and which direction each moves depends on the sign of the variation. For $f_I > 1$, the survival term suppresses W below unity (a larger cross section makes it less likely the particle survived to the interaction point), while the interaction term pushes W above unity (the channel that occurred is now more probable). For $f_I < 1$ both directions reverse: the survival term enhances W above unity, and the interaction term suppresses it below. A particle whose track does not end in a hadronic inelastic interaction receives only the survival term, and so is suppressed for $f_I > 1$ and enhanced for $f_I < 1$.

This is also the reason the trajectory granularity matters. The survival term is a product over steps, so replacing many short steps by one long step changes the result: the exponent is evaluated at a single momentum rather than following the particle as it slows down. Section III A returns to this point, and Section V B quantifies it.

B. The Interaction-Step Weight in `G4Reweighter`

The SI systematic uncertainty implementation presented in this note follows the approach described in [2]. Initial `G4Reweighter` implementation included the bare scale factor alone,

$$W_I^{(\text{as distributed})} = f_I, \tag{5}$$

without the rate-ratio factor $(\sigma_i/\sigma_i')\,(1 - e^{-\Delta L_i \sigma_i'})/(1 - e^{-\Delta L_i \sigma_i})$ correction.

Therefore, we modified `G4Reweighter` definition by extending it to the form now defined in Eq. 3, that applies the survival exponential of Eq. 2 at every step, including the interacting step. Ref. [2] specifies that a step receives exactly one of the two weight forms: a non-interacting step gets W_S (Eq. 2), and an interacting step gets W_I (Eq. 3). Eq. 3 is the form used consistently throughout this note.

Section V B 1 verifies Eq. 3 against a hand calculation on a single track, and Figure 2 compares the resulting weight distributions of Eq. 3 (green-dotted histogram) to those from Eq. 5 (red histogram).

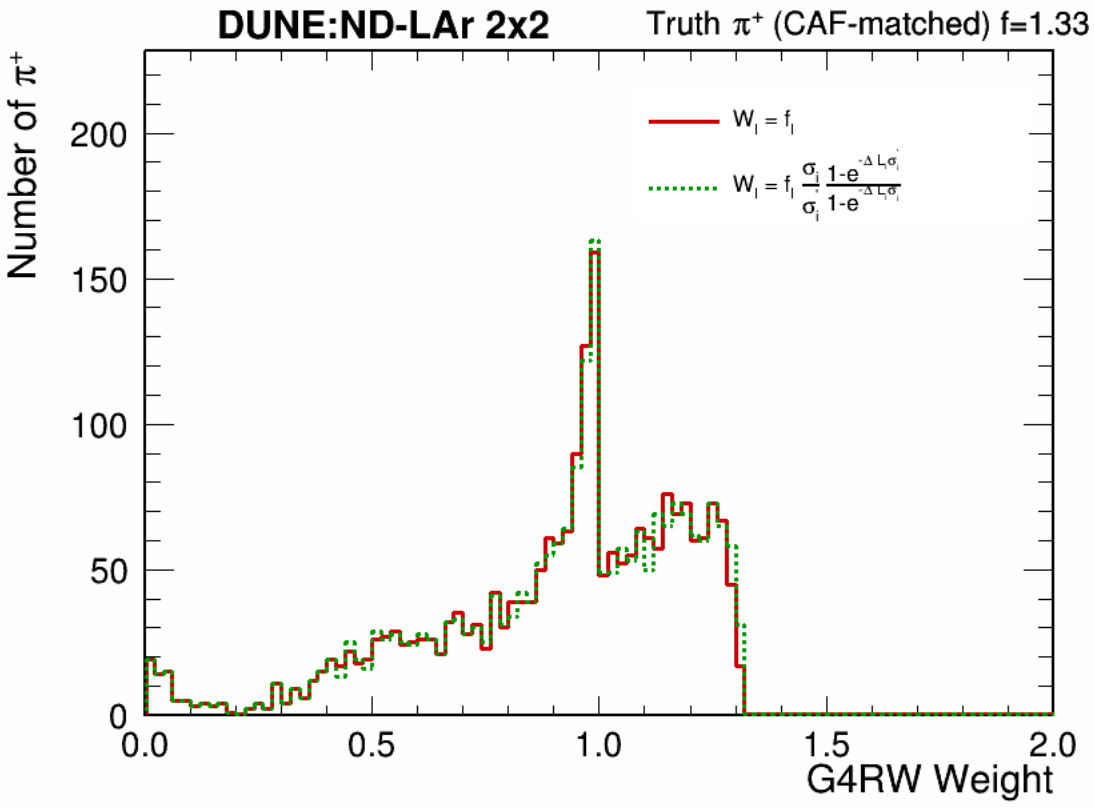


FIG. 2. G4RW weight distribution for CAF-matched truth π^+ at $f = 1.33$, comparing $W_I = f_I$ (Eq. 5) against W_I as given by Eq. 3. The shapes agree closely.

C. Exclusive Channel Decomposition

G4RW can scale the total inelastic cross section as a single number, or scale individual exclusive channels. The latter requires knowing how `Geant4` distributes inelastic interactions among final states as a function of momentum. That information is supplied by the exclusive channel fraction files, referred to throughout as fraction (aka "fracs") files.

For charged pions the channels are absorption (`abs`), charge exchange (`cex`), double charge exchange (`dcex`), quasielastic scattering (`inel`), and pion production (`prod`). The fractions are dimensionless and normalized at every momentum point:

$$f_{\mathrm{abs}}(p) + f_{\mathrm{cex}}(p) + f_{\mathrm{dcex}}(p) + f_{\mathrm{inel}}(p) + f_{\mathrm{prod}}(p) = 1. \tag{6}$$

The exclusive cross section for channel c is recovered at runtime as

$$\sigma_c(p) = f_c(p)\,\sigma_I(p), \tag{7}$$

where $\sigma_I(p)$ is the total inelastic term from `Geant4`'s internal tables. The fracs file therefore stores only the decomposition, not the cross section itself.

For nucleons the same scheme is used with channels defined by counting outgoing nucleons: no outgoing nucleons (`0n0p`), one proton only (`0n1p`), one neutron only (`1n0p`), one of each (`1n1p`), and all remaining multi-nucleon topologies (`Other`, labelled `MnMp` in the figures). This decomposition is not part of the standard distribution and was produced for this analysis, as presented in Section III B.

The current analysis scales the total inelastic cross section rather than individual channels; the fracs file is what would make per-channel variation possible.

III. REQUIRED INPUTS

Running G4RW requires three inputs: EDepSim files carrying full `Geant4` trajectories, fracs files giving the exclusive channel decomposition, and a FHiCL configuration file specifying what to vary and by how much. Each is described below. The first two required work specific to this analysis.

A. EDepSim Files with Full Trajectories

G4RW evaluates Eq. 2 step by step, reading the position and momentum at each `Geant4` transport step. EDepSim does not store them by default. Its `SelectTrajectoryPoints` routine, in `EDepSimPersistencyManager.cc`, prunes trajectory points geometrically, keeping points at volume boundaries and the track endpoints. For G4RW this is not enough: a hadron crossing a uniform argon volume is reduced to as few as two points, and the product over steps in Eq. 2 collapses to as little as a single interval.

1. *The patch*

Throughout this note, *patch* refers to the EDepSim sample produced with full trajectory points retained, as distinguished from the default *unpatched* sample, in which `SelectTrajectoryPoints` prunes most points away. The fix is a bypass we inserted into `SelectTrajectoryPoints` ahead of the pruning logic. For the species of interest, every stored trajectory point is kept and the routine returns before the geometric pruning runs; behaviour for all other particles is unchanged. The bypass condition is

```
if (std::abs(pdg) == 211 || pdg == 2212 ||
    pdg == 2112 || std::abs(pdg) == 321) {
    for (int i = 0; i < g4Traj->GetPointEntries(); ++i)
        selected.push_back(i);   // keep every step
    return;
}
```

covering $\pi^{\pm}$ ($\pm$211), protons (2212), neutrons (2112), and $K^{\pm}$ ($\pm$321).

2. Validation

EDepSim was recompiled with this fix and used to regenerate the full simulated 1000-file sample. A diagnostic pass over the whole sample gives the trajectory point counts in Table I. All six species show the full-trajectory signature. The minimum of 2 points that appears for every species corresponds to tracks that stop after a single step and is expected.

TABLE I. Trajectory points per track in the patched sample.

Particle	PDG	Average points	Min	Max
π^+	+211	47.3	2	1159
π^-	−211	44.2	2	1194
proton	+2212	9.1	2	7482
neutron	+2112	54.4	2	1080
K^+	+321	54.1	2	939
K^-	−321	50.5	2	869

Table II shows the effect on a single track. The unpatched listing carries two points and one step of 6.83 cm; the patched listing carries seven points resolving the same 6.83 cm into six steps, with the momentum falling from 1180 to 1116 MeV along the way. The total path length is unchanged, which is the expected behaviour: the patch changes only how finely the trajectory is sampled.

TABLE II. Stored trajectory points for a single π^+ track before (Unpatched) and after (Patched), listing position, step length, momentum, and `Geant4` process and subprocess codes at each point. The total path length is 6.83 cm in both cases; the patch changes only how finely that length is sampled.

File=0000000 Spill(entry)=3 RunId=0 EventId=3 TrackId=6 nPoints=7

	step	x [mm]	y [mm]	z [mm]	ΔL [cm]	$\lvert p\rvert$ [MeV]	proc	subproc
Unpatched	0	−1006.6	1167.6	2723.6	0.00	1180.2	2	22
	1	−1004.8	1152.6	2790.2	6.83	0.0	4	121
Patched	0	−1006.6	1167.6	2723.6	0.00	1180.2	2	22
	1	−1006.0	1163.5	2742.3	1.92	1158.2	2	2
	2	−1005.8	1161.4	2751.6	0.95	1149.4	1	91
	3	−1005.6	1159.4	2760.2	0.89	1149.4	1	91
	4	−1005.1	1155.2	2778.8	1.90	1123.7	2	2
	5	−1005.0	1154.6	2781.3	0.26	1115.9	2	2
	6	−1004.8	1152.6	2790.2	0.91	0.0	4	121

3. Choice of EDepSim sample

The EDepSim sample used for weight production must satisfy three conditions at once: it must reproduce the track lengths seen in the CAF, its event numbering must match the CAF interaction identifier, and it must carry full trajectories. Five candidate samples were compared — the official MiniRun5 production plus four produced at Argonne — differing in geometry (`Merged2x2MINERvA_v4_withRock` versus `Merged2x2MINERvA_v4_noRock`), patch status, and GENIE input set. Their properties are summarised in Table III.

TABLE III. EDepSim samples compared for G4RW weight production. All ANL samples use only `genie.nu` GTRAC files (argon interactions); the official sample additionally includes `genie.rock`. "Patched" refers to preservation of intermediate `Geant4` trajectory points.

#	Sample	Geometry	Patched	GENIE input	Matches CAF
1	Official unpatched	`withRock`	No	`nu` + `rock`	Yes
2	ANL unpatched	`noRock`	No	`nu` only	No
3	ANL patched	`noRock`	Yes	`nu` only	No
4	ANL unpatched	`withRock`	No	`nu` only	Yes
5	ANL patched	`withRock`	Yes	`nu` only	Yes

The geometry choice determines where a hadron's trajectory ends: in `withRock`, a hadron leaving the argon immediately enters rock and interacts there, consistent with the CAF; in `noRock`, the surrounding material is vacuum, so the same hadron travels considerably further before interacting, which is why the `noRock` samples fail to match the CAF. Sample 5 (`patched_withRock_genieNu`) is the only one satisfying all three conditions and is used for all weight production reported here.

A fraction of CAF hadrons originate in the upstream rock rather than in argon. These are present in the official sample but absent from the ANL samples, which use `genie.nu` only. They have no entry in the weight files, receive the default weight $w = 1.0$, and contribute no systematic shift — the correct treatment, since they are not argon interactions.

B. Exclusive Channel Fraction Files

The fracs file supplies the channel fraction $f_c(p)$ of Eq. 7, the fraction of the total inelastic cross section carried by channel c at momentum p, one object per channel. The standard `dune_pardata` distribution [13] provides a full five-channel decomposition for π^+, covering 11–9001 MeV/c; this file is used unmodified. We generated fracs files for π^-, protons, and neutrons using the `G4Cascade` application, extending to the same upper momentum of 9001 MeV/c. A hadron with momentum above this range has no per-channel decomposition available and receives a weight of $w = 1.00$; Figure 3 shows the exclusive channel fractions for all four species.

To produce fracs files, `G4Cascade` fires 10,000 particles into a liquid argon target at each of a set of momentum points, classifies the final state of each inelastic interaction, and writes the resulting channel fractions. The neutron configuration is shown below; the proton case differs only in `Type` and `Outfile`, and the π^- case additionally sets `NPi0_Cex: true` with 150 MeV thresholds on charged pion secondaries, matching the settings used for the official π^+ file so the two are directly comparable.

```
#include "g4rw_material.fcl"
#include "g4rw_physlist.fcl"
Type:        2112
NCascades:   10000
Range:       [11., 9001.]
NDivisions:  880
Outfile:     ".../G4RW/G4Cascade/LAr_N_cascade_0_9000.root"
Material:    @local::LAr
NPi0_Cex:    false
Thresholds:  []
Physlist:    @local::Physlist_default
```

C. FHiCL Configuration

The FHiCL file tells G4RW what to vary. It specifies the target material, the path to the fracs file, the overall scale factor (`Weight`), and a `ParameterSet` listing the channels to vary. Each entry in the `ParameterSet` carries the channel name (`Cut`), the momentum range over which the variation applies (`Range`), the nominal value (always 1.0, since a scale factor of 1 reproduces the unmodified `Geant4` cross section), and the prior width (`Sigma`). A sample π^+ configuration is shown below.

```
Weight: 0.67
```

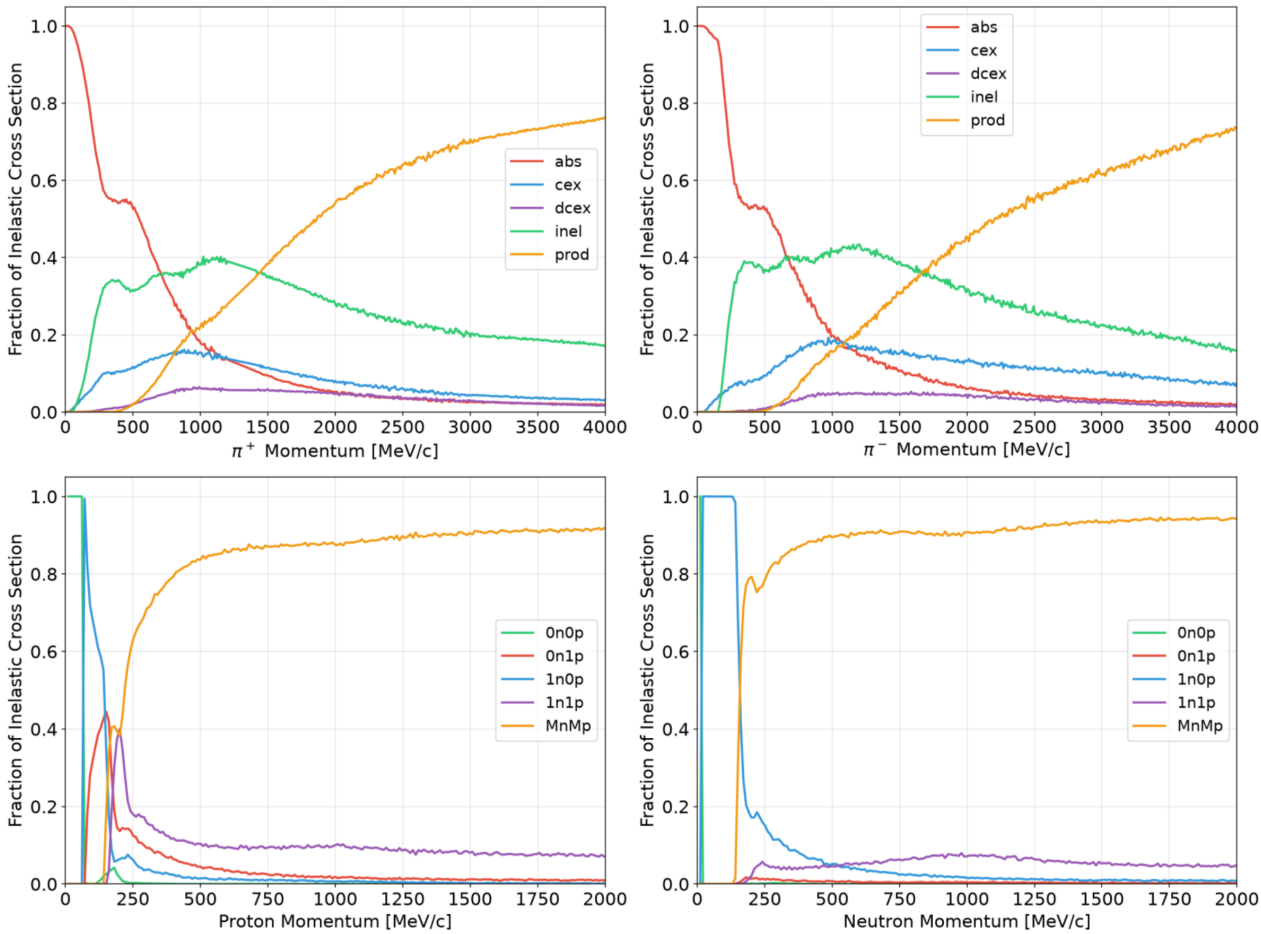


FIG. 3. Exclusive channel fractions as a function of momentum in liquid argon, computed with `G4Cascade` by firing 10,000 particles into argon at each momentum point and classifying the outgoing final state. Top: π^+ (left, official `dune_pardata` file) and π^- (right, generated at ANL), with channels `abs`, `cex`, `dcex`, `inel`, `prod`. Bottom: proton (left) and neutron (right), both generated at ANL, with nucleon-counting channels `0n0p`, `0n1p`, `1n0p`, `1n1p`, and `MnMp`. Fractions sum to unity at every momentum point in all four cases. For visual clarity, the plotted momentum range is truncated below the full fracs-file range of 11–9001 MeV/c: pions are drawn up to 4000 MeV/c and protons and neutrons up to 2000 MeV/c.

```
Material: {
  Name: "liquidArgon"
  Density: 1.400
  Components: [
    { Z: 18
      Mass: 39.948
      Fraction: 1. }
  ]
}
FracsFile: "Geant4ReweightInput/LAr_PiPlus_cascade_0_9000.root"
ParameterSet: [
  { Cut: "reac"
    Name: "fReac"
    Range: [0., 2000.]
    Nominal: 1.0
    Sigma: 0.67 }
]
```

The single `Cut: "reac"` entry scales the total inelastic cross section, and is the only cross section varied; with no `Cut: "elast"` entry present, $\sigma'_E = \sigma_E$ and the elastic contributions cancel in Eq. 2.

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The scale factor's prior follows the parameterization described in Refs. [14, 15]: a ±33% width around the nominal value of 1.0, giving 0.67 as the downward variation and 1.33 as the upward one. This width was adopted for current studies, motivated by the spread between `Geant4` predictions and external π–A scattering data, shown in Figure 4.

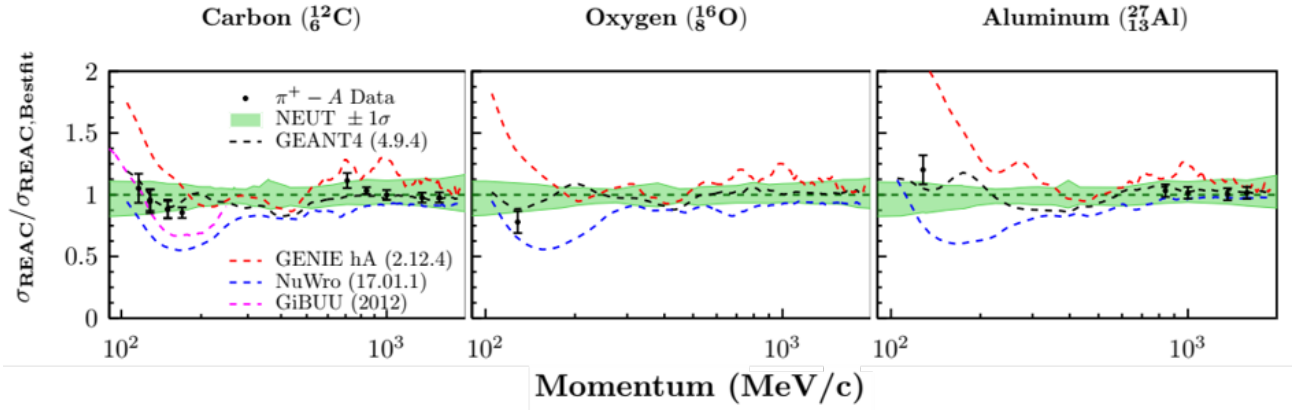


FIG. 4. Ratio of the reactive cross section to its best-fit value for π^+ on carbon, oxygen, and aluminium, comparing external scattering data against `Geant4` and several neutrino-generator cascade models, from Ref. [15]. The spread between models at a given momentum is what motivates the ±33% prior adopted here for argon.

IV. EXTENDING G4RW TO SIX HADRON SPECIES

Extending G4RW to the species relevant for the 2×2 and ND-LAr touched three largely independent components: the EDepSim trajectory patch, which determines whether a species has usable trajectories at all (Section III A); the fracs file, which supplies the channel decomposition (Section III B); and the `Geant4Reweight` source, which determines whether a species has a reweighter class with exclusive channels (Section IV A). A fourth component, the 2×2 weight-production executables and their FHiCL configurations, is what actually produces weight files (Section V). Table IV gives the state of each hadron species across each of these components.

TABLE IV. Coverage by species and by component. "Native" means the capability was already present in the distributed framework and required no change here. "Weight production" refers to the component that produces per-particle weight trees. Kaons carry the EDepSim patch as groundwork only, so that trajectories exist if kaon reweighting is taken up later.

Species	EDepSim patch	G4RW code	Fracs file	Weight production
π^+	Added	Native	Native (`dune_pardata`)	Done
π^-	Added	Native	Generated at ANL	Done
proton	Added	Modified	Generated at ANL	Done
neutron	Added	Modified	Generated at ANL	Done
K^+	Added	TBD	TBD	TBD
K^-	Added	TBD	TBD	TBD

A. `Geant4Reweight` Source Changes

Giving nucleons an exclusive channel decomposition required modifying two source files. Both changes mirror a pattern already present for pions. The π^- required no source changes at all, since `G4PiMinusReweighter.cc` already implements all five exclusive channels natively.

a. Reweighter classes. The `G4ProtonReweighter.cc` and `G4NeutronReweighter.cc` were each changed in two places. The constructor channel list was changed from the single placeholder channel to the five nucleon-counting channels:

```
// before
: G4Reweighter(..., {"total"}, ...)
// after
: G4Reweighter(..., {"0n0p", "0n1p", "1n0p", "1n1p", "Other"}, ...)
```

and `GetInteractionSubtype()`, which returned a hardcoded string, was replaced by logic that classifies each inelastic interaction by counting secondary nucleons:

```
std::string G4NeutronReweighter::GetInteractionSubtype(
    const G4ReweightTraj & theTraj) {
  int nProtons  = theTraj.HasChild(2212).size();
  int nNeutrons = theTraj.HasChild(2112).size();
  if      (nNeutrons == 0 && nProtons == 0) return "0n0p";
  else if (nNeutrons == 0 && nProtons == 1) return "0n1p";
  else if (nNeutrons == 1 && nProtons == 0) return "1n0p";
  else if (nNeutrons == 1 && nProtons == 1) return "1n1p";
  else                                      return "Other";
}
```

The proton version is identical apart from the class name.

b. Parameter maker. `G4ReweightParameterMaker.cc` defines which channel names are accepted for each species. The proton (2212) and neutron (2112) cases were changed from {`"total"`, `"reac"`} to the five exclusive channels plus `"reac"`. Retaining `reac` is necessary: it is the parameter used when scaling the total inelastic cross section.

V. WEIGHT PRODUCTION AND VALIDATION

A. Scale Factors and the Weight-Production Executable

Although the SI cross-section channels and its scale factors are implemented as independent, the current production varies all cross-section channels by a same scale factor. The five SI cross-section variations are performed, using the following scale factors applied in each of five single passes over each file:

$$f \in \{0.34,\ 0.67,\ 1.33,\ 1.66,\ 2.00\}. \tag{8}$$

The $\pm 33\%$ pair ($f = 0.67, 1.33$) is the physically motivated one (Section III C). The remaining three ($f = 0.34, 1.66, 2.00$, corresponding to $\pm 66\%$ and $+100\%$) are included to exercise the machinery across a wider range.

Weights are produced by a C++ executable that runs over the patched EDepSim trajectories, computes the per-track G4RW weight for each of the five scale factors, and writes the result to a `TTree` named `g4rwEvt` with one entry per weighted particle, carrying the identifiers needed to match a weight back to a reconstructed particle (see Table V).

TABLE V. Branches of the `g4rwEvt` output tree.

Branch	Type	Meaning
`file_idx`	`int`	File index parsed from the input filename
`spill_id`	`int`	EDepSim spill index
`run_id`	`int`	EDepSim `RunId`
`event_id`	`int`	`RunId` $\times 10^6 +$ `EventId`
`track_id`	`int`	`Geant4` track ID
`pdg`	`int`	PDG code of the reweighted particle
`weight`	`double`	G4RW weight, one branch per scale factor
`len`	`double`	Total track length (cm)

B. Single-Track Validation

Before looking at distributions, it is worth following an example where the machinery is applied on a single track, where every step can be inspected. Figure 5 does this for a π^+ with 45 recorded trajectory points spanning 14.4 cm

and ending in a hadron-inelastic interaction.

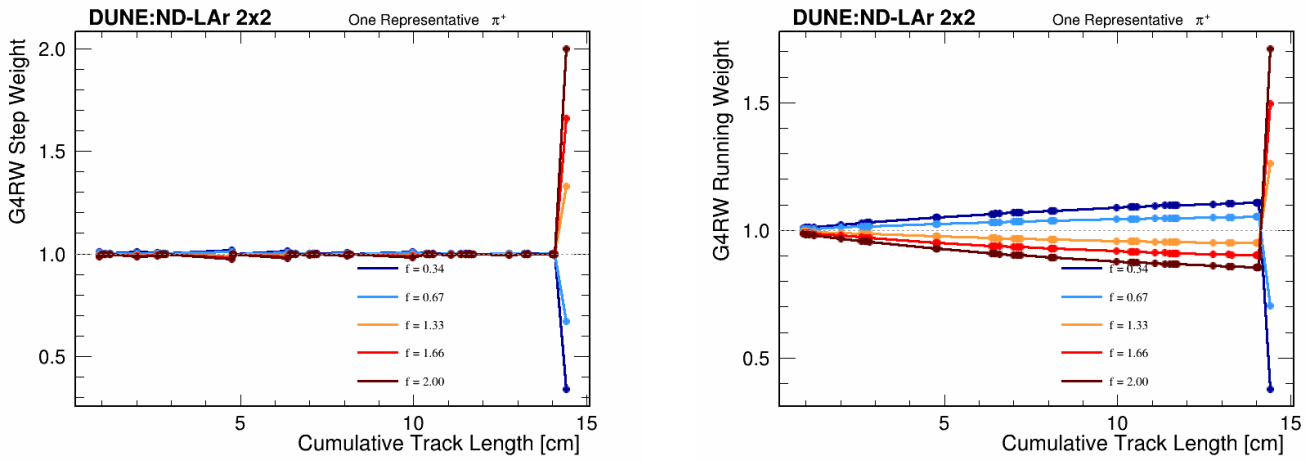


FIG. 5. Per-step weight (left) and cumulative running weight (right) along a single π^+ track of 45 trajectory points and 14.4 cm, for each of the five scale factors. The per-step weight stays close to unity at every non-interacting step, because a step length is a small fraction of the mean free path. The final step carries the interaction term of Eq. 3 and is not constrained to lie near unity, which produces the sharp excursion at the end of the track.

The shape explains a feature that recurs throughout the distributions below. At every non-interacting step the survival ratio of Eq. 2 deviates from unity only slightly, because $\Delta L_i \sigma_i$ is small. For $f < 1$ survival is slightly *more* likely at each step, so the running weight drifts above unity along the track; for $f > 1$ it drifts below. The final step then applies the interaction term, which scales with f itself and pulls the weight in the opposite direction. The curves therefore cross near the end of the track.

1. *Verifying the final step by hand*

Table VI works through the final step of that track explicitly, and reproduces the saved production weight. This is the check that confirms the Eq. 3 implementation described in Section II B is correct.

TABLE VI. Hand evaluation of Eq. 3 at the final step of the track in Figure 5 (File 0000000, RunId 0, EventId 7, TrackId 3, PDG 211), at $f = 1.33$. The result reproduces the saved production weight.

Quantity	Value
Running weight after 44 steps, W_S	0.950046
Last step length, ΔL_{45}	0.345925 cm
σ_I^{nom} at last step ($p \approx 84.84$ MeV/c)	0.003982 cm^{-1}
σ_I' at last step $= f\,\sigma_I^{\text{nom}}$	$1.33 \times 0.003982 = 0.005296$ cm^{-1}
σ_E^{nom} at last step	0.002484 cm^{-1}
$\sigma = \sigma_I + \sigma_E$	$0.003982 + 0.002484 = 0.006466$ cm^{-1}
$\sigma' = \sigma_I' + \sigma_E$	$0.005296 + 0.002484 = 0.007780$ cm^{-1}
W_I (Eq. 3)	$1.33 \times \frac{0.006466}{0.007780} \times \frac{0.00268766}{0.002234219} = 1.329698$
Final weight	$0.950046 \times 1.329698 = 1.263274$

C. Weight Distributions

Although we the SI systematic uncertainty is already implemented for π^+, π^-, protons, and neutrons, we illustrate the results with detailed π^+ and proton examples.

Figure 6 shows the weight distributions for truth π^+ and truth protons from matched interactions (from CAFs), one curve per scale factor. Taking $f = 1.33$ as representative, two features stand out in both species: a distinct

two-peak accumulation near $w = 1.00$ and $w = 1.33$, and a tail of weights extending down towards zero.

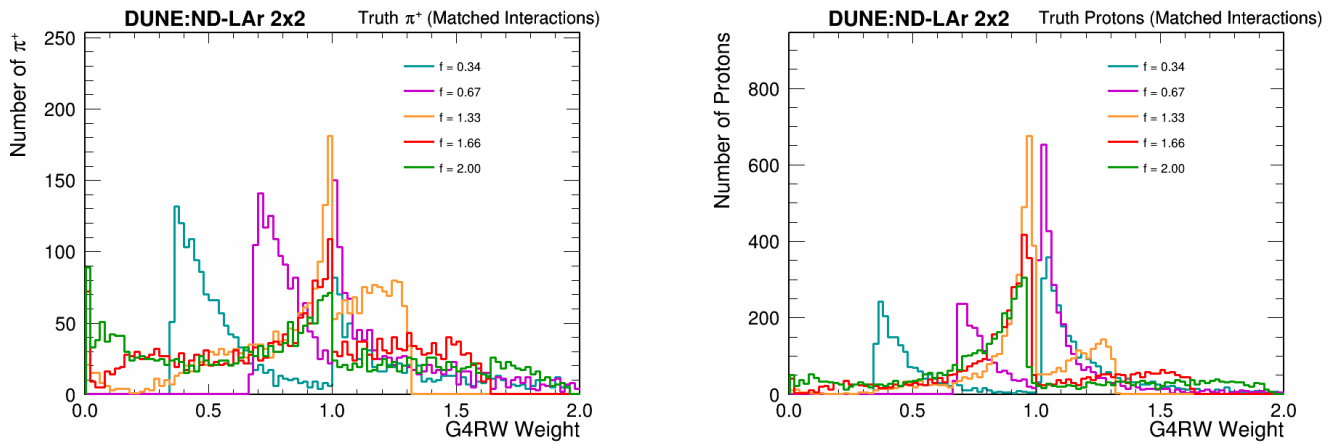


FIG. 6. G4RW weight distribution from matched interactions in the patched sample, one curve per cross-section scale factor, for truth π^+ (left) and truth protons (right). All five curves share the same general structure: a population accumulating near $w = 1.00$, a second near $w = f$, and a tail extending down towards zero.

Plotting the weight against track length separates the two populations cleanly and explains both features directly. Figure 7 shows the two-dimensional π^+ and proton distributions at $f = 1.33$.

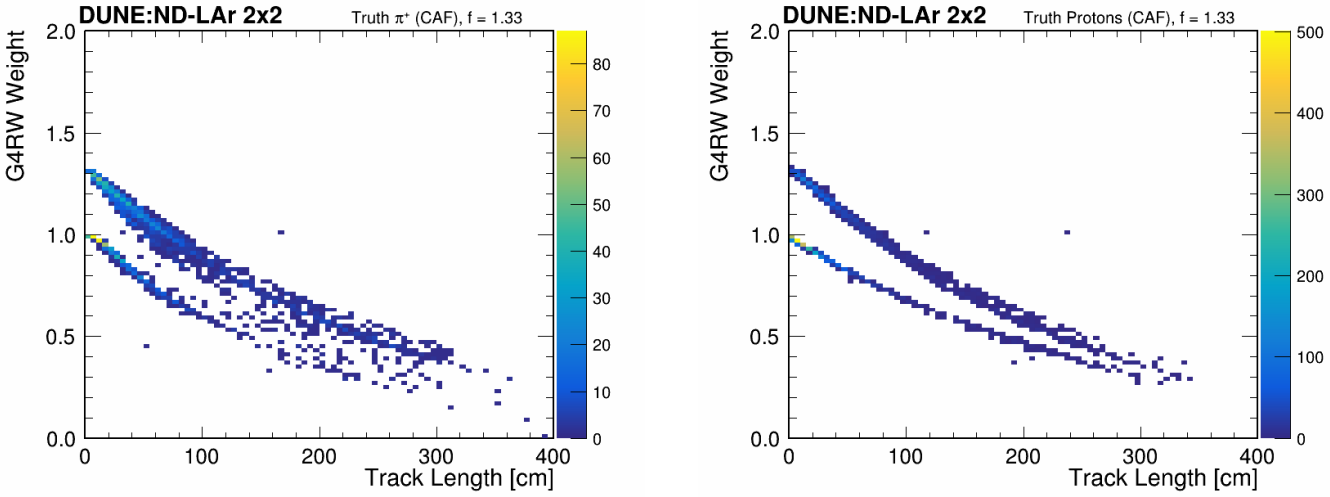


FIG. 7. Cumulative G4RW weight against truth track length at $f = 1.33$, for truth π^+ (left) and truth protons (right). Two bands are visible in both particle cases, converging near $w = 1.00$ and $w = 1.33$ at short track length and falling together as the track lengthens. The lowest weights come from the longest tracks, which are predominantly those leaving the fiducial volume. The scattered entries sitting at $w = 1.00$ away from either band are hadrons with momentum above the upper edge of the fracs file.

The two bands are the two terms of Eq. 4. A track whose last recorded point is tagged as a hadronic inelastic interaction receives both W_S and W_I, and lands on the upper band, which starts near $w = f$ at zero length. A track whose last point is tagged as anything else receives only W_S, and lands on the lower band, which starts at $w = 1.00$. Both bands then fall with track length as the survival term accumulates.

This can be tested directly by splitting on the `Geant4` process and subprocess code recorded at each track's last point. Table VII gives the composition and Figures 8 and 9 show the resulting split.

The difference between the two species in Table VII is worth noting. Protons end in a hadronic inelastic interaction only 41.8% of the time against 70.3% for pions, with the remainder stopping by ionisation. Fewer proton tracks therefore receive an interaction term, which is why the $w = 1.00$ peak is relatively more prominent for protons than for pions in Figure 6.

TABLE VII. Composition of the `Geant4` process/subprocess code at the last recorded trajectory point, for truth π^+ and truth protons. Only the hadron-inelastic category receives the interaction term of Eq. 3. Pions have a decay component that protons do not.

Geant4 Code (`proc/subproc`)	Meaning	π^+	proton
4/121	hadron inelastic	70.3%	41.8%
2/22	electromagnetic/scintillation	26.4%	58.2%
6/201	decay	3.3%	—

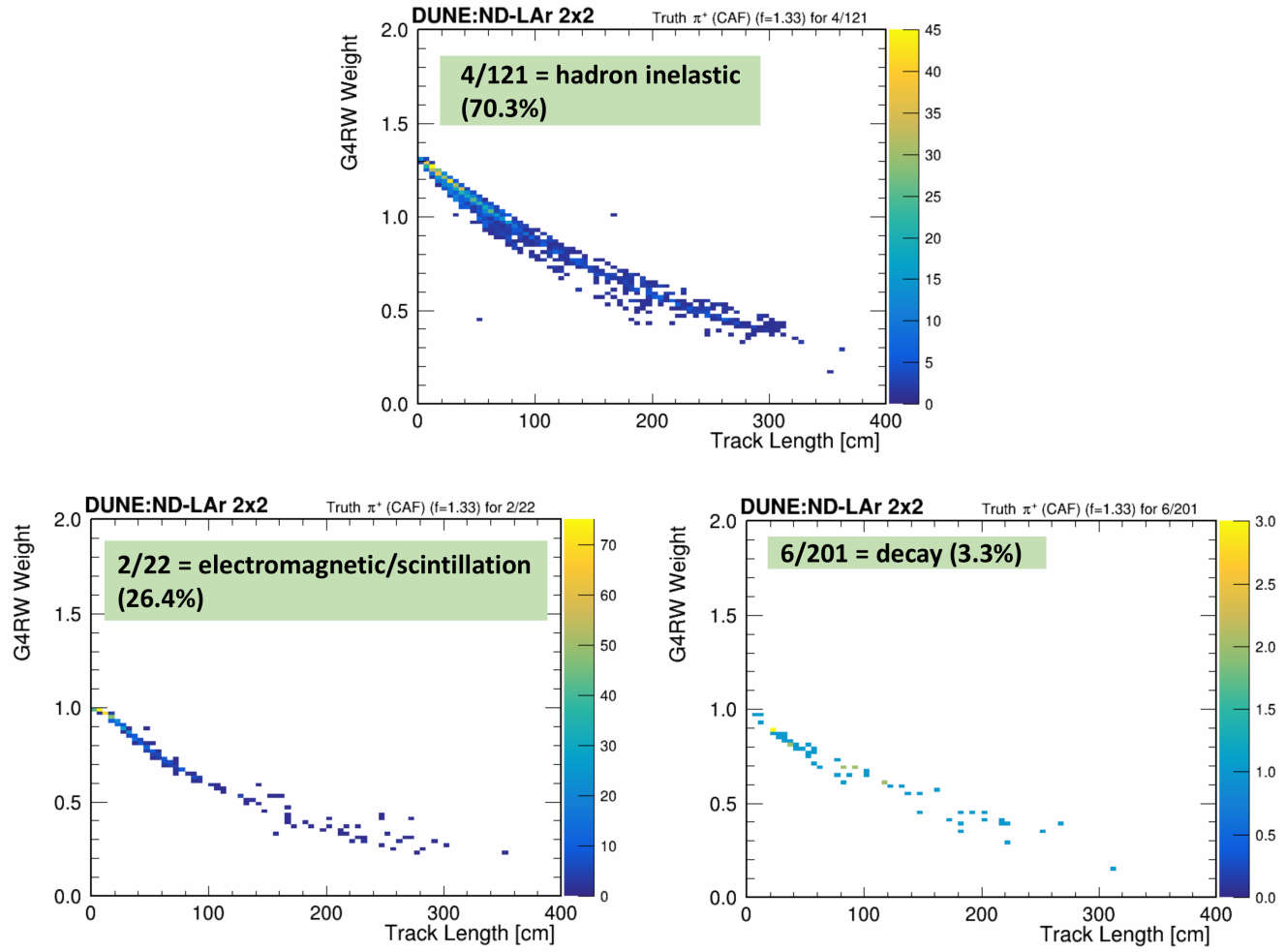


FIG. 8. Weight against track length for truth π^+ at $f = 1.33$, split by the `Geant4` process/subprocess code at the last recorded point: hadron inelastic (4/121, top), electromagnetic/scintillation (2/22, bottom left), and decay (6/201, bottom right). Only the first begins at $w = f$; the other two begin at $w = 1.00$, confirming that the upper band is exactly the population receiving the interaction term.

Some scattered entries at $w = 1.00$ in Figure 7 are those hadrons whose momentum exceeds the upper edge of the fracs file ($|p|$ >9001 MeV/c; Section III B), where the channel decomposition is undefined and no scaling is applied.

VI. SI WEIGHT PARAMETERIZATION VS. TRUE TRACK LENGTH

The weight interface described so far is a per-hadron lookup: each weight is retrieved using the key (`cafFileIdx`, `nu_id`, `G4ID`), and exists only for hadrons that were generated and produced a matching entry in the weight file (Section VII A).

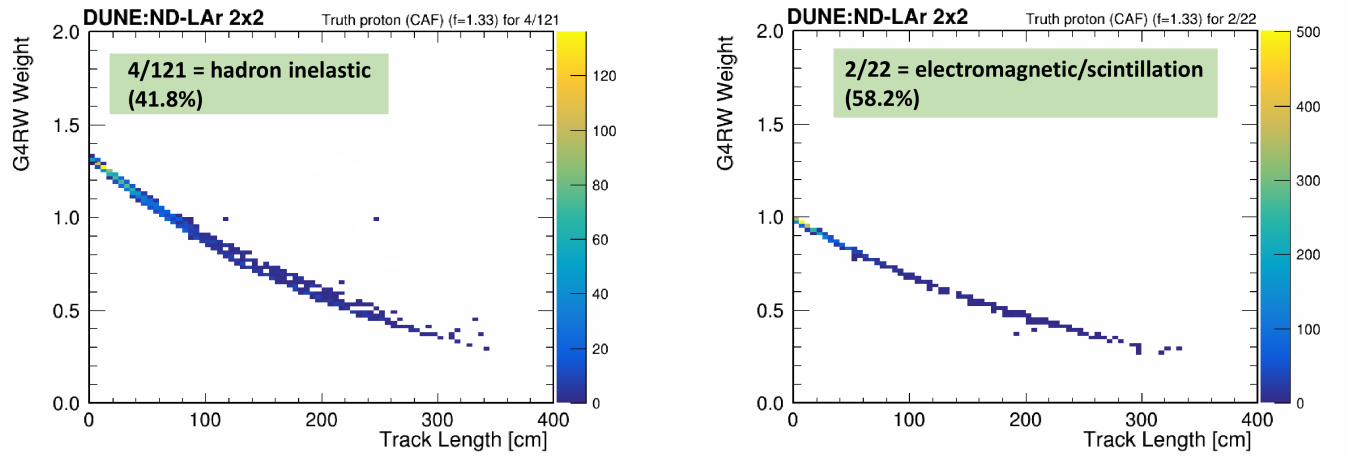


FIG. 9. Weight against track length for truth protons at $f = 1.33$, split by the `Geant4` process/subprocess code at the last recorded point. Protons divide between hadron inelastic (4/121) and electromagnetic/scintillation (2/22) only.

This framework is complete, and its use is demonstrated in Section VIII (Figures 12, 13, and 14). We extended it with a continuous parameterization of the weight as a function of the true track length L, a polynomial $w(L) = \sum_{k=0}^{n} c_k\, L^k$ fit to the scatter of per-hadron weights already produced by the framework. The parameterization gives the weight at any point in true or reconstructed phase space; in particular, for a hadron that leaves the fiducial volume, it can be evaluated at the true length accumulated up to the point where the hadron exits the detector.

The scatter separates into two populations with distinct weight-length trends, identified using the Geant4 process/subprocess tag at each hadron's last recorded point: hadrons whose last point is tagged hadronic-inelastic (4/121), versus hadrons whose last point is tagged something other than 4/121. We examine this separately for protons and π^+, using the same $[5, 290]$ cm fit range for both populations of each species.

We tested polynomial fits of order 3 through 7 for both populations of each species. Figures 10 and 11, together with Tables VIII and IX, show the fitted weight at $L = 200$ cm for each order; across all four cases (proton and π^+, each tagged 4/121 or tagged other than 4/121), pol3 is the simplest solution, with a minimal fit uncertainty when compared to other polynomial fits tested.

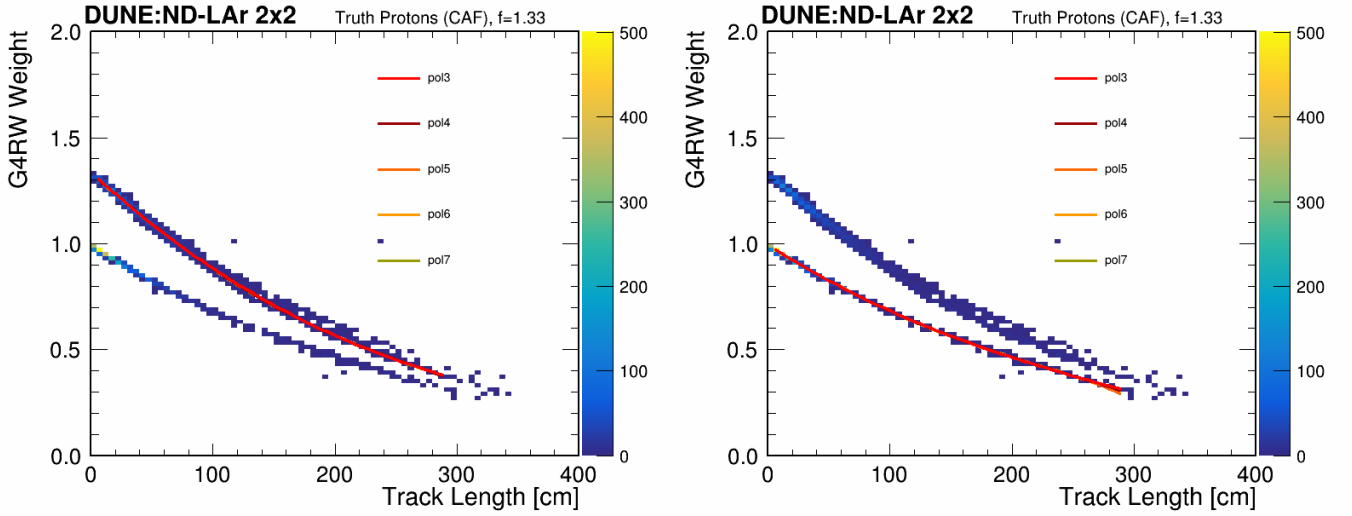


FIG. 10. Preliminary polynomial fits (order 3–7) to G4RW weight ($f = 1.33$) vs. true track length for truth protons, separated by the Geant4 process/subprocess tag at each track's last recorded trajectory point: last point tagged hadronic-inelastic, 4/121 (left), and last point tagged something other than 4/121 (right).

A π^+ whose last point is *not* tagged hadronic-inelastic can reach that state via two distinct processes: electromagnetic/scintillation (2/22) or decay (6/201). The lower-population curve in Figure 11 is drawn only up to $L = 240$ cm:

TABLE VIII. Fitted G4RW weight at $L = 200$ cm, by polynomial order, for protons whose last trajectory point is tagged hadronic-inelastic (4/121) versus tagged something other than 4/121.

Order	$w(L = 200\,\text{cm})$, 4/121	$w(L = 200\,\text{cm})$, other than 4/121
pol3	0.5663 ± 0.0012	0.4631 ± 0.0009
pol4	0.5660 ± 0.0012	0.4638 ± 0.0010
pol5	0.5661 ± 0.0016	0.4636 ± 0.0010
pol6	0.5670 ± 0.0019	0.4644 ± 0.0012
pol7	0.5668 ± 0.0023	0.4644 ± 0.0012

beyond that point per-length-bin statistics decrease sharply (several bins with only 2–5 tracks), and the higher-order fits (pol6, pol7) become visibly unstable there. The fit itself still uses the full $[5, 290]$ cm range; only the drawn range is truncated.

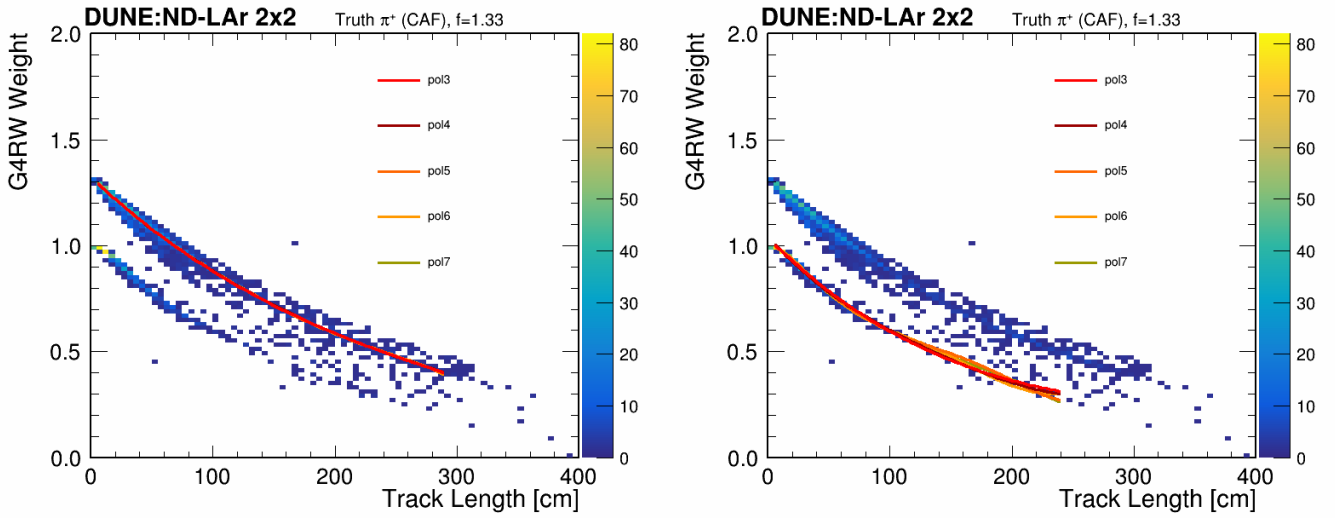


FIG. 11. Preliminary polynomial fits (order 3–7) to G4RW weight ($f = 1.33$) vs. true track length for truth π^+, separated by the Geant4 process/subprocess tag at each track's last recorded trajectory point: last point tagged hadronic-inelastic, 4/121 (left), and last point tagged something other than 4/121 (right, drawn over $[5, 240]$ cm, see text).

TABLE IX. Fitted G4RW weight at $L = 200$ cm, by polynomial order, for π^+ whose last trajectory point is tagged hadronic-inelastic (4/121) versus tagged something other than 4/121.

Order	$w(L = 200\,\text{cm})$, 4/121	$w(L = 200\,\text{cm})$, other than 4/121
pol3	0.5861 ± 0.0017	0.3645 ± 0.0033
pol4	0.5862 ± 0.0017	0.3539 ± 0.0038
pol5	0.5846 ± 0.0021	0.3651 ± 0.0039
pol6	0.5840 ± 0.0023	0.3387 ± 0.0049
pol7	0.5841 ± 0.0026	0.3678 ± 0.0063

Given this stability, the pol3 fit is adopted as the parameterization,

$$w(L) = c_0 + c_1 L + c_2 L^2 + c_3 L^3, \tag{9}$$

with L in cm. Tables X and XI list the fitted coefficients, with propagated errors, for protons and π^+ respectively.

TABLE X. Fitted pol3 coefficients (Eq. 9) for protons, with propagated errors.

Population	c_0	c_1	c_2	c_3
4/121	$1.3371(\pm0.0013)$	$-5.2975(\pm0.0541)\times10^{-3}$	$7.9967(\pm0.4791)\times10^{-6}$	$-3.8995(\pm1.1614)\times10^{-9}$
other than 4/121	$0.9941(\pm0.0006)$	$-3.6969(\pm0.0307)\times10^{-3}$	$6.4614(\pm0.3297)\times10^{-6}$	$-6.2622(\pm0.9220)\times10^{-9}$

TABLE XI. Fitted pol3 coefficients (Eq. 9) for π^+, with propagated errors.

Population	c_0	c_1	c_2	c_3
4/121	$1.3241(\pm0.0022)$	$-5.4381(\pm0.0851)\times10^{-3}$	$1.0925(\pm0.0762)\times10^{-5}$	$-1.0915(\pm0.1904)\times10^{-8}$
other than 4/121	$1.0356(\pm0.0017)$	$-5.6875(\pm0.0922)\times10^{-3}$	$1.4203(\pm0.1049)\times10^{-5}$	$-1.2708(\pm0.2847)\times10^{-8}$

VII. PROPAGATION TO TRUTH AND RECONSTRUCTED OBSERVABLES

A. Matching Weights to Reconstructed and Truth-Level Particles

The SI weights are computed on MiniRun5 EDepSim files, but the observables, both reconstructed and truth-level, are read from MiniRun6.5 CAF files. Three files must therefore be linked: the EDepSim file, the weight file, and the CAF file.

Files correspond one to one by index. Within a file, interactions are matched by a unique identifier built from the EDepSim run and event numbers,

$$\texttt{event_id} = \texttt{RunId} \times 10^6 + \texttt{EventId}, \tag{10}$$

where `RunId` encodes which of the ten GENIE sub-files inside a merged EDepSim file the interaction came from and `EventId` counts interactions within that sub-file. This identifier survives the full chain (GENIE → EDepSim → `larnd-sim` → FLOW → SPINE → CAF) and appears in the CAF as `nu.id`. Individual particles are matched by the `Geant4` track identifier, stored as `prim.G4ID` in the CAF.

The three-way key is summarised in Table XII. Because the key is built entirely from truth-level identifiers (`G4ID`, `nu.id`), it resolves identically regardless of whether the resulting weight is applied to a particle's true or reconstructed kinematics; Section VIII A uses this to apply the same SI weight map to both the reconstructed and truth-level track length of a common set of backtracked proton candidates. A worked implementation of both the reconstructed and truth-level fills is available in the multiplicity analysis code [16], commit `ab109f6` (reconstructed fill: lines 3300–3324; truth fill: lines 3328–3347).

TABLE XII. The three-way lookup key linking CAF, weight, and EDepSim files.

File	File index	Interaction ID	Particle ID
CAF	loop index f	`nu.id`	`prim.G4ID`
Weights	`file_idx`	`event_id`	`track_id`
EDepSim	filename index	$\texttt{RunId}\times10^6+\texttt{EventId}$	`TrackId`

A particle with no entry in the weight map receives $w = 1.0$ and contributes no shift.

B. Building the Systematic Band

Weights are applied when filling histograms. Each entry falls in one bin, and for each observable this gives a nominal and a shifted histogram,

$$h_i^{\mathrm{nom}} = \sum_{\text{entries in bin } i} 1, \qquad h_i^{\mathrm{wt}} = \sum_{\text{entries in bin } i} w. \tag{11}$$

Since the current production supplies one weight per particle per scale factor, the shift in bin i is

$$\delta_i = h_i^{\mathrm{wt}} - h_i^{\mathrm{nom}}, \tag{12}$$

and the uncertainty band in the figures is $\sigma_i = |\delta_i|$, using only that bin's own shift.

The band construction above depends only on each track's weight w, not on how it was obtained, so it applies equally whether w is evaluated from the fitted parameterization of Section VI or a bare weight in the G4RW output ROOT files (Section VII A).

VIII. ANALYSIS EXAMPLES WITH INCORPORATED SI, GENIE, AND DETECTOR SYSTEMATICS

In this section, we incorporate systematics uncertainties from SI, nusystematics, and detector response in two example analyses; in both, the SI weight is the bare weight from the G4RW output ROOT files (Section VII A), not the fitted parameterization of Section VI.

A. Example of SI Systematics in Reconstructed and Truth-Level Proton Track Length

As we study charged hadron multiplicity, we need to understand the individual primary particle distribution. To that end we looked at the proton sample, and used it as a test bed for incorporating the SI systematic uncertainty, as well as all other systematic contributions as described in this section.

Figure 12 shows the SI band propagated to the reconstructed proton track-length distribution, alongside the GENIE and detector response uncertainties, and their combination. The proton candidate selection is the one documented in Ref. [10]; the GENIE interaction systematics are produced with the `nd_nusyst` framework for MiniRun6.5, as described in Ref. [17], and the detector response systematics are the current set from Ref. [18]. Simulation is normalized to data (Sandbox v11) [19] throughout.

Since the SI weight is defined at the level of the true `Geant4` trajectory, it can equally be applied to the true kinematics of the same backtracked candidates. Of the 3354 reconstructed proton candidates entering Fig. 12, truth-backtracking identifies 2257 (67.3%) as true protons, 622 (18.5%) as π^-, 276 (8.2%) as π^+, and 199 (5.9%) as other species or without a truth match. Each candidate is weighted with the SI weights of its true species (proton, π^+ or π^- weight file); the last group is not reweighted ($w = 1$). Figure 13 shows the SI band for the 2257-candidate proton sample (top) and for the 1097 non-proton candidates (bottom) at both reconstructed and true track length, using raw MC counts with no normalization to data. Since these figures are not normalized to data, their bands should be compared with the band in Fig. 12 through their fractional width rather than their absolute size.

B. Example of Systematic Uncertainty Propagation in an older Multiplicity Analysis

In this subsection, we incorporate the SI systematic uncertainties, along with nusystematics and detector response uncertainties, to an older version of multiplicity analysis [20, 21] that is still useful to illustrate the individual systematic uncertainty contributions.

Figure 14 shows all three systematic ingredients — SI from G4RW, GENIE interaction systematics [17], and detector response systematics [18] — applied to the charged-track multiplicity distribution. The SI systematic band is negligible in the first bin (multiplicity 1), since this bin is dominated by muons, which are insensitive to secondary hadronic interactions.

IX. SOFTWARE AND FILE LOCATIONS

This section lists the software and file locations needed to reproduce the work described in this note. All paths below are on NERSC. `$CFS` abbreviates `/global/cfs/cdirs/dune/users/mazam`.

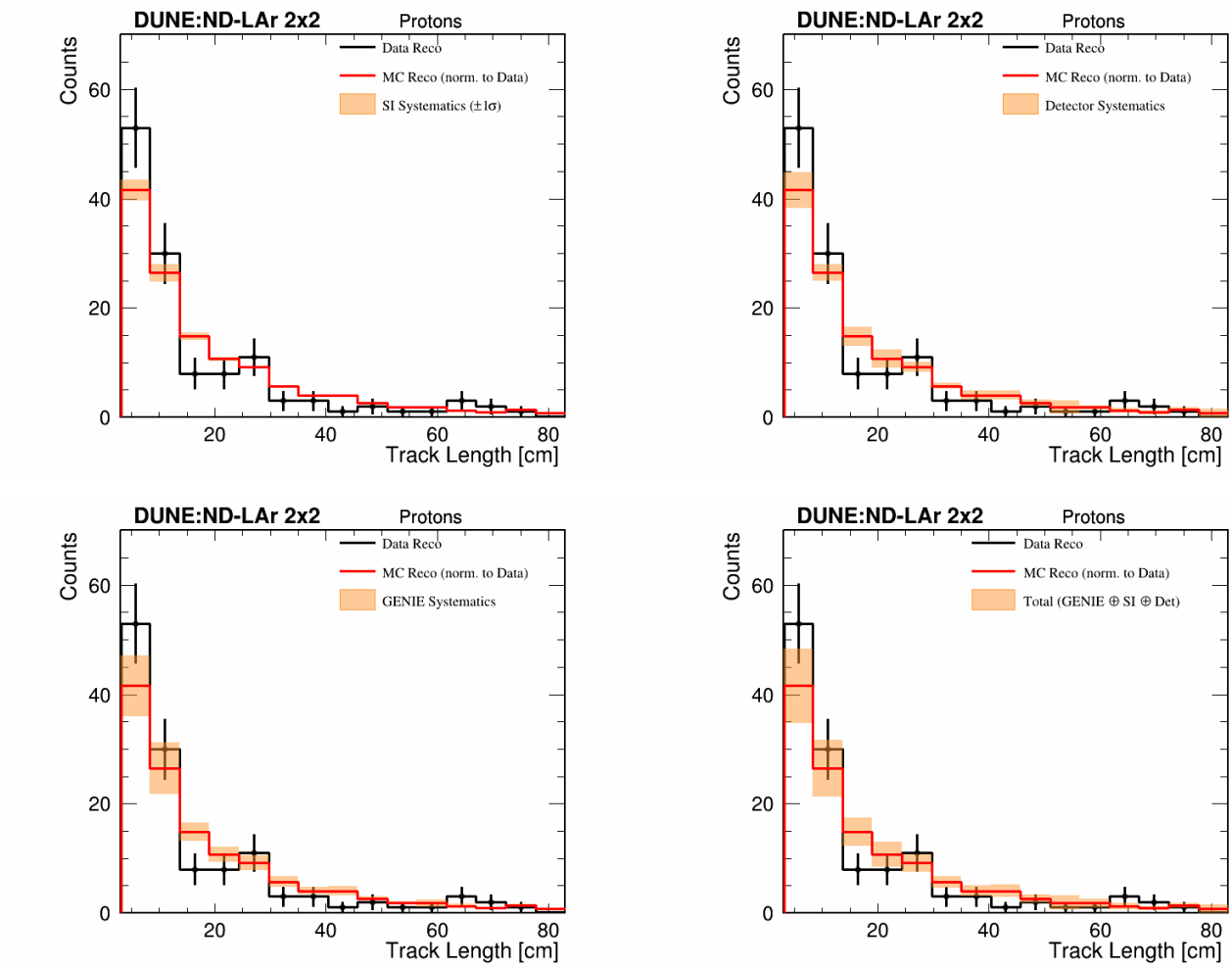


FIG. 12. Reconstructed proton track length. Black: data. Red: simulation, normalized to data. Shaded bands: systematic uncertainty. Panels show the secondary-interaction band from G4RW at $\pm 1\sigma$, the detector response band, the GENIE band, and their quadrature sum.

TABLE XIII. Sample, weight, and input file locations.

Item	Path
Patched EDepSim ($\pi^{\pm}/p/n/K^{\pm}$)	`$CFS/EdepSim_G4RW_nu/patched_withRock_genieNu/`
G4RW weight outputs	`$CFS/G4RW/output_weights/patched_withRock_genieNu_eq3p10/`
Fracs files (π^{-}, p, n)	`$CFS/G4RW/G4Cascade/LAr_*_cascade_*.root`
Fracs file (π^{+}, official)	`$DUNE_PARDATA_DIR/Geant4ReweightInput/LAr_PiPlus_cascade_0_9000.root`
README files	`$CFS/softwares/README`

TABLE XIV. Software locations and the patched source files.

Item	Path
2×2 G4RW pipeline	`$CFS/softwares/2x2_g4rw/`
Custom `Geant4Reweight` build	`$CFS/softwares/geant4reweight_custom/`
Patch archive	`$CFS/softwares/patches/`
EDepSim patch source	`$CFS/EdepSim_G4RW/patch/EDepSimPersistencyManager.cc`
EDepSim production script	`$CFS/EdepSim_G4RW/batch_scripts/run_g4rw_edepsim_batched.sh`

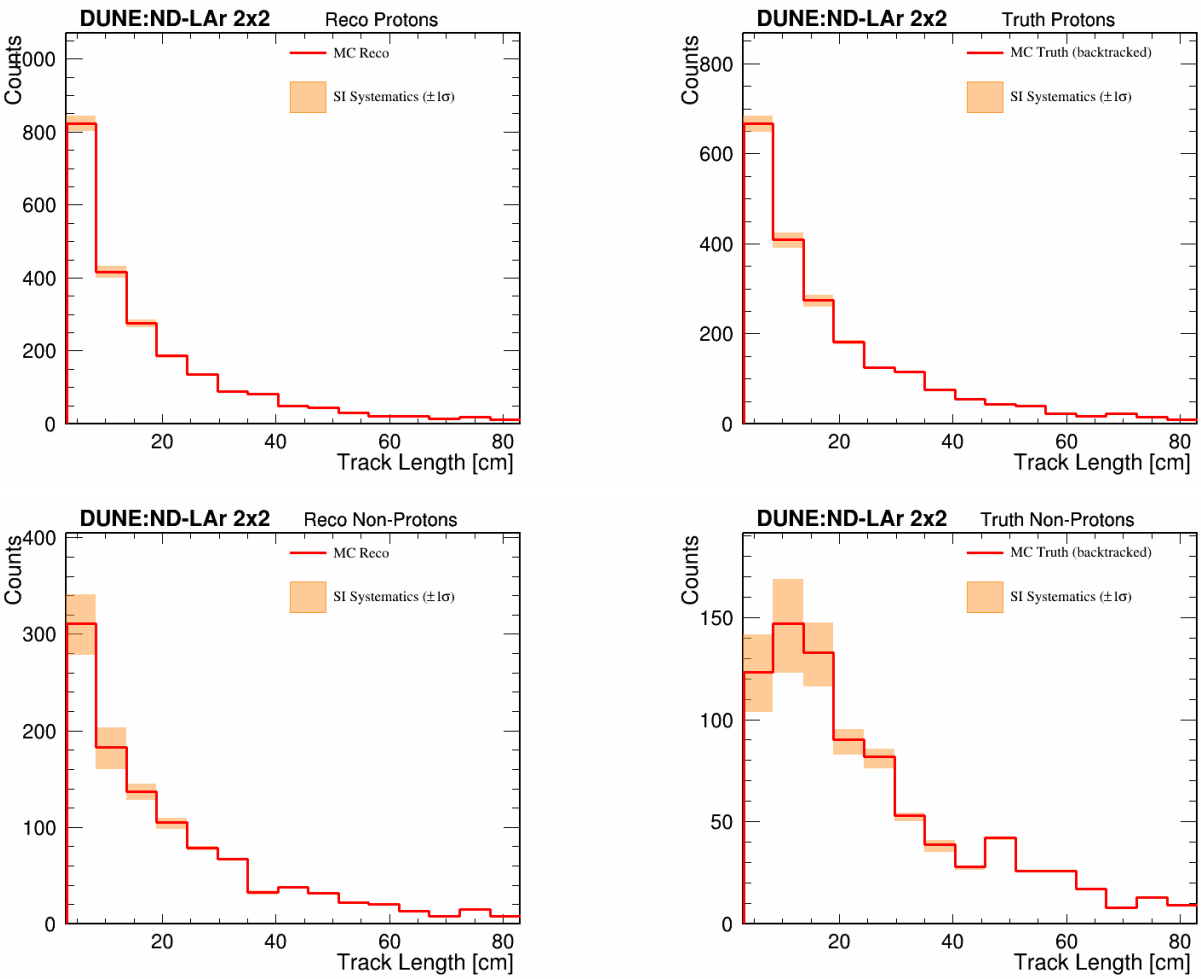


FIG. 13. Reconstructed (left) and truth-level (right) track length, in raw MC counts not normalized to data, with the SI systematic band at $\pm 1\sigma$. Top: the 2257 correctly-backtracked proton candidates. Bottom: the 1097 non-proton candidates at reconstructed level, and the 969 with a truth match (622 π^-, 276 π^+, 71 other species) at truth level; the remaining 128 have no truth match and appear only in the reconstructed-level panel.

TABLE XV. Software versions used. The two `boost` versions are required at runtime: the locally built libraries link against `v1_80_0` while the UPS dependency closure links against `v1_82_0`, and both must be on `LD_LIBRARY_PATH`.

Package	Version	Note
`2x2_G4RW`	commit `9bf2556`	
`edep-sim`	tag `3.2.0.2x2.1`	patched, see Sec. III A
`2x2_sim`	commit `7c62186`	geometry and run scripts
`Geant4Reweight`	`v01_21_02`	patched, see Sec. IV A
`Geant4`	`v4_11_2_p02`	
ROOT	`v6_28_12`	
`cetmodules`	`v3_24_01`	
`boost`	`v1_80_0`, `v1_82_0`	both needed at runtime
`dune_pardata`	`v01_84_00`	official π^+ fracs file

Two containers are used. The `Geant4Reweight` and 2×2 G4RW builds run in `fermilab/fnal-wn-sl7:latest`, which supplies the Scientific Linux 7 environment and the DUNE UPS stack. The EDepSim build and production run in `mjkramer/sim2x2:ndlar011`; this container is required rather than optional, because the 2×2 GDML geometry contains `G4Torus` shapes that fail with a mismatched `Geant4` version.

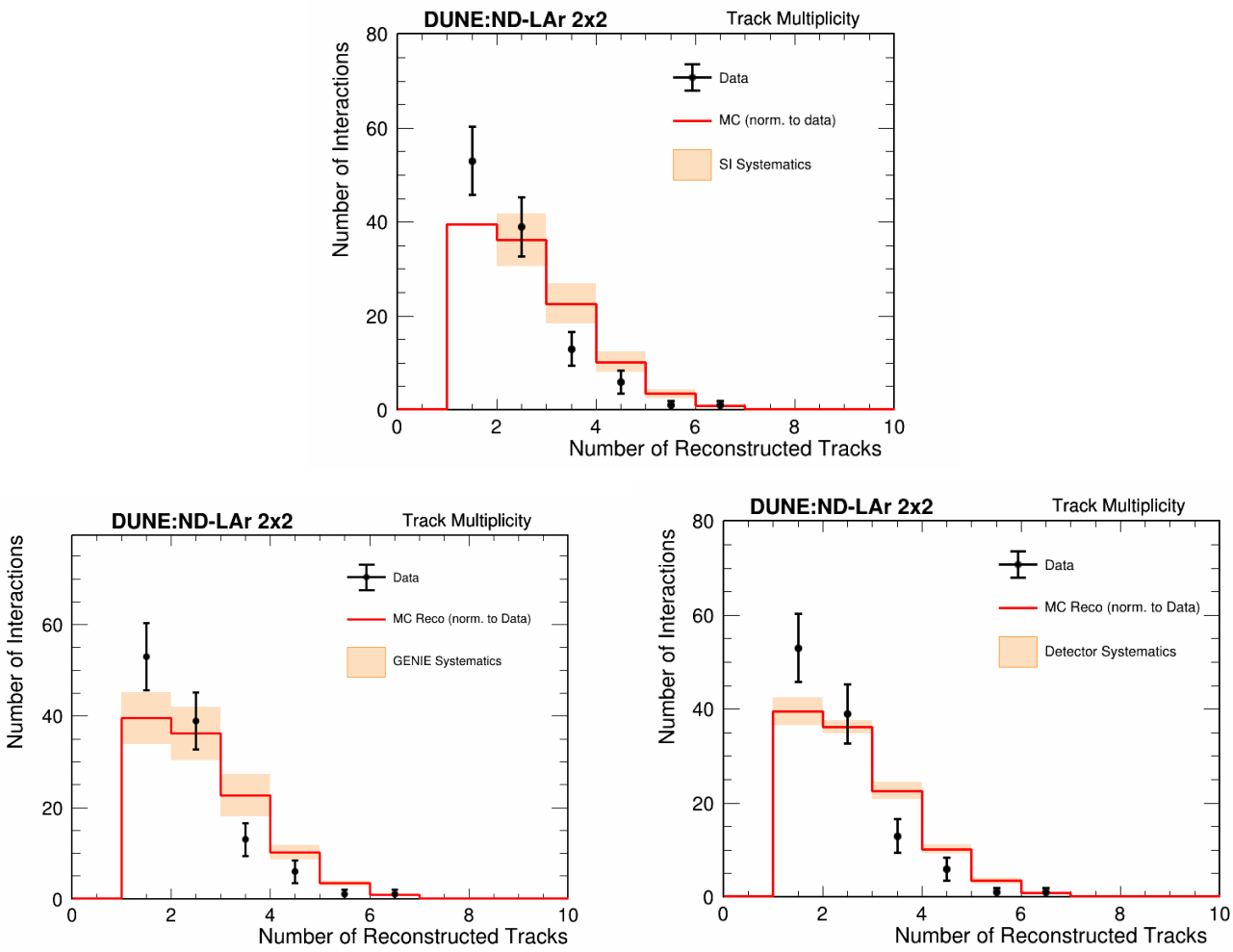


FIG. 14. Reconstructed charged-track multiplicity with the secondary-interaction band from G4RW (top), the GENIE interaction systematic band (bottom left), and the detector response systematic band (bottom right). Black: data. Red: simulation, normalized to data.

Currently, the entire codebase lives on NERSC and we plan to migration it to a GitHub repository, so that the weight production chain is reproducible outside the NERSC environment. With these locations established, the remainder of this section gives a short recipe for using the resulting weights in a CAF analysis.

A. Applying G4RW Weights in a CAF Analysis

The following steps show how to look up the SI weight for a particle in a CAF analysis.

Step 1: Use the G4RW weight files (one file per hadron species) from following path:

```
$CFS/G4RW/output\_weights/patched\_withRock\_genieNu\_eq3p10/
```

Step 2: Load the `g4rwEvt` tree of the required species into a lookup table keyed by (`file_idx`, `event_id`, `track_id`), with the weight as the value.

Step 3: For each track in the CAF analysis, build the key (`cafFileIdx`, `nu.id`, `prim.G4ID`) as defined in Table XII, where `cafFileIdx` is the index of the CAF file in the processing order. In the multiplicity analysis code this is done as:

```
// interaction ID (nu.id)
```

```
g4rw_nu_id    = (long long)sr->mc.nu[biggestMatchIndex].id;
// index of the current file in the CAF chain
g4rw_fileIdx = caf_chain->GetTreeNumber();
// Geant4 track ID of the matched particle (prim.G4ID)
g4rw_G4ID     = sr->mc.nu[biggestMatchIndex].prim[partNumber].G4ID;
```

Step 4: Look up this key in the table from Step 2. If an entry is found, the corresponding weight is read from it; if not, the nominal weight $w = 1.0$ is assigned to this track. Steps 2 and 4 are implemented as:

```
std::map<std::tuple<int,int,int>, std::array<double,2>> wmap;

TFile* f = TFile::Open(weightFile);
TTree* t = (TTree*)f->Get("g4rwEvt");
int file_idx, event_id, track_id;
double w_1p33, w_0p67;
t->SetBranchAddress("file_idx",    &file_idx);
t->SetBranchAddress("event_id",    &event_id);
t->SetBranchAddress("track_id",    &track_id);
t->SetBranchAddress("weight_1p33", &w_1p33);
t->SetBranchAddress("weight_0p67", &w_0p67);
for (long i = 0; i < t->GetEntries(); i++) {
    t->GetEntry(i);
    wmap[{file_idx, event_id, track_id}] = {w_1p33, w_0p67};
}

// per track: nominal weight unless an entry is found
std::array<double,2> w = {1.0, 1.0};
auto it = wmap.find({cafFileIdx, (int)nu_id, G4ID});
if (it != wmap.end()) w = it->second;
```

Step 5: Apply the weight to that track's kinematics (true or reconstructed). A worked implementation is available in the multiplicity analysis code [16].

X. SUMMARY

This note has documented the extension of G4RW to the 2×2 Demonstrator: patching EDepSim to retain full trajectories of six hadron species ($\pi^{\pm}, p, n, K^{\pm}$), extending the G4RW framework and its fracs files to four hadronic species ($\pi^{\pm}, p, n$), implementing the full interaction-step weight formula. Currently, kaons have the EDepSim patch only; no reweighter class work, fracs file, or weight production is done for them; the patch is groundwork so that the trajectories exist if kaon reweighting is taken up later. The SI systematics are then propagated to reconstructed proton (Fig. 12) and multiplicity distributions (Fig.14) alongside GENIE and detector systematics. We also parameterized the weight as a continuous function of true track length for protons and π^{+}. Software and file locations for reproducing this work are given in Section IX.

With the secondary-interaction uncertainty now available for these observables, the next phase of this work moves from developing the systematic itself to using it in downstream physics analysis: unfolding the reconstructed distributions, optimizing the hadron-multiplicity selection, and other analysis steps that depend on having this systematic in hand. This study will be revisited as improved cross-section measurements become available to inform the underlying uncertainties.

[1] DeepLearnPhysics Collaboration, SPINE: Scalable particle imaging with neural embeddings, https://github.com/DeepLearnPhysics/spine (2026), version 0.2.3 used in the MiniRun6.5 production.
[2] J. Calcutt, C. Thorpe, K. Mahn, and L. Fields, JINST **16** (08), P08042, arXiv:2105.01744.
[3] NuSystematics, `nusystematics`: Neutrino interaction systematic weight calculators, https://github.com/NuSystematics/nusystematics (2026), accessed August 2026.
[4] S. Agostinelli *et al.*, Nucl. Instrum. Meth. A **506**, 250 (2003).
[5] DUNE Production, MiniRun5: ND-LAr 2x2 simulation production, https://wiki.dunescience.org/wiki/MiniRun_5 (2026), accessed August 2026.

[6] DUNE Production, MiniRun6: ND-LAr 2x2 simulation production, https://wiki.dunescience.org/wiki/MiniRun_6 (2026), accessed August 2026.

[7] Z. Djurcic, M. B. Azam, *et al.*, Geant4 ReWeighting and interaction systematics, DUNE-doc-36481-v3, https://docs.dunescience.org/cgi-bin/sso/ShowDocument?docid=36481 (2026), 5 March 2026.

[8] Z. Djurcic, M. B. Azam, *et al.*, Geant4 ReWeighting systematics, DUNE-doc-36847-v2, https://docs.dunescience.org/cgi-bin/sso/ShowDocument?docid=36847 (2026), 23 April 2026.

[9] Z. Djurcic, M. B. Azam, *et al.*, Analysis update: Geant4 ReWeighting systematics, ND Prototypes Analysis Workshop, 14–16 September 2026 (2026).

[10] M. B. Azam and Z. Djurcic, NDLAr-2x2 proton sample distributions, DUNE-doc-37081, https://docs.dunescience.org/cgi-bin/sso/ShowDocument?docid=37081 (2026).

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